

Genome-wide analysis and environmental response profiling of SOT family genes in rice (*Oryza sativa*)

Rongjun Chen · Yunyun Jiang, Jiali Dong · Xin Zhang · Hongbo Xiao · Zhengjun Xu · Xiaoling Gao

Received: 20 March 2012 / Accepted: 27 May 2012 / Published online: 10 October 2012
© The Genetics Society of Korea and Springer 2012

Abstract

Sulphotransferase (SOT) catalyses the transfer of a sulphonate group from 3'-phosphoadenosine 5'-phosphosulphate (PAPS) to an appropriate hydroxyl group of various substrates with the parallel formation of 3'-phosphoadenosine 5'-phosphate (PAP). Although several *SOTs* have been identified and characterized in mammalian, their role in plant is still unclear. In this study, we report genome-wide comprehensive expression analysis of 35 putative SOT genes in rice. The 35 *OsSOTs* were tandemly arranged into six clusters. The phylogenetic analysis showed that there were 7 subfamilies of *OsSOTs* and 11 putatively conserved motifs. Six *OsSOTs* might be pseudogenes, 25 have the two motifs which were involved in PAPS binding regions I and IV. Microarray data indicated that all the *OsSOTs* were expressed almost at the same level but with different patterns: most *OsSOTs* were expressed exclusively in stigma and ovary and induced by IAA and BAP, several genes were induced by tZ and DMSO and 11 *OsSOTs* were response to abiotic stress. Further analysis showed that these 11 genes contained *cis*-regulatory elements responding to abiotic stresses.

Keywords Expression patterns; Functional analysis; Microarray; *Oryza sativa*; Phylogenetic analysis; Sulphotransferase

Introduction

Sulphotransferases (SOT) catalyze the transfer of a sulphonate group from 3'-phosphoadenosine 5'-phosphosulphate (PAPS) to an appropriate hydroxyl group of various substrates with the parallel formation of 3'-phosphoadenosine 5'-phosphate (PAP). They have conserved amino acid motives, which seem

to be involved in PAPS binding (regions I and IV) (Baek et al., 2010; Cinta and Luc, 2008; Klein and Papenbrock, 2004, 2008). And SOTs catalyze the sulphonation of a wide range of compounds, producing sulphate esters and sulphate conjugates (Gidda and Varin, 2006; Hempel et al., 2007; Nowell and Falany, 2006). The SOT proteins were classified on the basis of their affinity for different classes of substrates. One group of SOT proteins, mainly membrane-associated, accepts substrates as macromolecules, such as proteins and peptides, and glucosaminoglycans (Ananvoranich et al., 1994). The second group, usually soluble proteins, accepts substrates as small organic molecules, such as flavonoids, steroids, and xenobiotics with diverse chemical structures. It is hypothesized that sulphonation, via SOTs, affects the biological activity of certain compounds, thereby modulating physiological processes such as growth, development, and adaptation to stress (Marsolais et al., 2000; Varin et al., 1997; Yang et al., 1999). There were a number of sulfated metabolites in the plant organism, such as brassinosteroids, coumarins, flavonoids, gibberellic acids, glucosinolates (GLs), phenolic acids, steroid, sulphatise esters such as choline-*O*-sulfate, and terpenoids, which might be sulfated by SOT proteins. However, based on the above-mentioned criteria, a number of other eukaryotic SOT proteins, whose substrate specificity was elucidated, could not be classified in the existing scheme. Either the proteins showed less sequence identity, although they possessed the same substrate specificity as other members of the subfamily, or they used a substrate chemically very different from the substrate typical for a particular subfamily, but showed high sequence identities (Sugahara et al., 2003). Thus, the classification scheme has to be adapted to the latest results with respect to substrate specificity (Cinta and Luc, 2008; Klein and Papenbrock, 2004, 2008).

Members of the SOT family have been found in most organisms investigated to date, except in archaea. Mammalian SOTs have been studied more than plant ones. Mammalian SOTs catalyze the sulphate conjugation of many hormones, neurotransmitters, drugs, and xenobiotic compounds (Cinta and Luc,

R. Chen (✉) · Y. Jiang · J. Dong · X. Zhang · H. Xiao · Z. Xu · X. Gao

Rice Research Institute of Sichuan Agricultural University, Chengdu, Sichuan 611130, China
e-mail: chenrj8@yahoo.com.cn

2008; Klein and Papenbrock, 2008). The flavonol 3- and 4'-SOTs from *Flaveria* species (Asteraceae), which catalyze the sulfation of flavonol aglycones and flavonol 3-sulfates, respectively, were the plant SOTs which cDNA clones were first isolated (Faulkner and Rubery, 1992).

Eighteen SOT had been identified, and according to their sequence similarity seven groups had been formed (Klein and Papenbrock, 2004). AtSOT10 /At2g14920 and AtSOT12/At2g03760 have a brassinosteroid activity; AtSOT8 and AtSOT9 did not display significant catalytic activity with brassinosteroids (Marsolais et al., 2007); Three SOT proteins, AtSOT16, 17, and 18 (Klein et al., 2006) were desulfo-(ds) Gl SOT enzymes (Baek et al., 2010; Cinta and Luc, 2008; Klein and Papenbrock, 2004, 2008; Nakamura et al., 2011). AtSOT15/At5g07010 exhibited strong substrate specificity for 11- and 12-hydroxyjasmonate (Gidda et al., 2003). Just one AtSOT, AtSOT5/At3g45070 was a flavonol activity found with strict specificity for position 7 of flavonols. It is thought that flavonoids might be involved in plant growth and development (Cinta and Luc, 2008; Gidda and Varin, 2006; Marsolais et al., 2000).

The position-specific flavonol SOT from *F. chloraefolia* were the first plant SOTs characterized. These enzymes exhibit strict specificity for position 3 of flavonol aglycones, 3'- and 4'- of flavonol 3-sulfates and 7 of flavonol 3, 3'- and 3, 4'-disulfates. The highly purified enzymes were active as monomers, did not require divalent cations for activity and had a similar molecular mass of 35 kDa. Despite the early discovery and abundance of flavonol sulfates in *Flaveria* species, their biological significance remains to be elucidated. One report suggests that sulfated flavonols might play a role in the regulation of polar auxin transport (Jacobs and Rubery, 1988). Flavonol aglycones, such as quercetin and kaempferol, were found to bind to the naphthylphthalamic acid receptor and thus inhibit polar auxin transport from the basal end of stem cells (Cinta and Luc, 2008; Faulkner and Rubery, 1992). In contrast, sulfated flavonols act as antagonists of quercetin and thus allow auxin efflux from tissues where it is produced (Faulkner and Rubery, 1992).

Only 17 of 21 SOTs were present on the 24 k Affymetrix chip and for many of them the absolute signal intensity is very low. The mRNA levels for most of the *Arabidopsis* SOTs characterized to date were low in plants grown at normal growth conditions (Gidda et al., 2003; Lacomme and Roby, 1996; Piotrowski et al., 2004). Similar to *Arabidopsis*, in *B. napus* the mRNA levels of the SOTs were very low under normal conditions but were increased upon treatments with salicylic acid and ethanol, again pointing out the potential function of these enzymes in the stress response (Marsolais et al., 2000; Rouleau et al., 1999).

In this study, we have reanalyzed the members of SOT family in rice based on complete genome sequence and annotation,

and a total of 35 putative SOT genes were identified. And a comprehensive expression analysis during various stages of development, hormone treatments, abiotic and biotic stress conditions, been performed.

Materials and Methods

Plant materials and growth conditions

The germinal seeds of cultivated rice Pei'ai 64S (*Oryza sativa* L.) were suspended in a sterile solution of 0.1% HgCl₂ for 10 min, washed 3 times using running water, immersed for 3 days under 25°C and changed water once a day, then were germinated and grown in distilled water at 37°C for 2–3 days. They were partly sown in pots that put in net room at Institute of Subtropical Agriculture of Chinese Academy of Science. Plants were divided into one control and two treatment groups. The control group was maintained under normal growth conditions, and the treatment groups were exposed to drought, heat and cold stresses. The leaf samples were from one treatment group at the five-leaf stage, another group plants were put in other pots which contained five ones, as the test materials of the booting and flowering stages. All the biological and technical replicates have been done three times.

Rice cold, heat and water-deficit treatment

For the drought tests the water was poured away from the basin, the treatment groups were put in scaffold to dry out meanwhile the control group was kept the water level. The leaves were harvested when they started curling after 16 hr. For the heat tests the treatment group was exposed to 45°C for 2 hr then plants were harvested; for the cold tests the treatment group was harvested after exposed to 4°C for 12 hr at seedling stage and to 12°C for treatment 16 hr at booting and flowering stages. All the rice were seedlings at in a U.S. Percival produced PGC15.5 artificial climate chamber, while the control group was in another chamber at 28°C. Test groups and control groups were in the dark.

Draw materials and Sample preparation

Four or five countdown second leaves were collected from treatment and control group, and four or five leaves were harvested which were not out of the young panicle or the middle of spiked out flower. The materials were cut into pieces, then ground into a dry powder with liquid nitrogen, and immediately divided into pre-installed 1.0 ml TRIzol extraction (Invitrogen) in 1.5 ml centrifugal tubes, about 100 mg each tube. We used low temperature marker pen to mark labels, tightly closed lid, then vigorously vortex to sure the samples were mixed with TRIzol extraction, seal the tubes with Parafilm wrap. The samples were stored at -70°C until required.

Total RNA isolation

Total RNA was isolated from the frozen samples using TRIzol. Extract method using TRIzol reagent (Invitrogen). The samples saved in -70°C were taken out, vortex, chloroform ($200\text{ }\mu\text{l}$) was added, vigorously shaken for 15 sec, and then centrifuged at $12,000 \times g$ for 15 min at 4°C . The upper layer was carefully removed from each tube, was transferred to another centrifuge tube, isopropanol ($500\text{ }\mu\text{l}$) was added, precipitated for at least 1 hr at -40°C , then centrifuged to separate the RNA. The RNA pellets were twice washed by 75% ethanol, air dried and dissolved in the appropriate volume of RNase-free water. The purity of RNA was determined by the A260/280 absorbance ratio (1.9–2.0). Isolated RNAs were stored at -70°C , after checking the purity and integrity of 18 S, 5 S and 28 S rRNA bands on 1.5% agarose gel.

Microarray

According to Affymetrix expression microarray experiments manual provided by GeneTech Biotechnology Limited Company, Shanghai, the following steps were: (1) Total RNA extraction and purification; (2) cDNA synthesis and purification; (3) transcription cRNA synthesis and cRNA purification in vitro; (4) cRNA fragmentation, hybridization solution preparation; (5) chip hybridization; (6) elution chip; (7) scan chips; (8) data analysis.

Search and analysis of SOT proteins in Arabidopsis and rice

Protein sequences of OsSOTs were acquired from a query search in the BLAST program at NCBI using the amino acid sequence of the cloned *OsSOT9*, GRAMENE and rice genomes (TIGR version 6.1) with the key word of sulfotransferase. All the AtSOT proteins were searching by GRAMENE (http://www.gramene.org/db/searches/quick_search_for=sulfotransferase). OsSOT motif scan was performed using SMART and PFAM Database searches with filter off. OsSOT genes were mapped on chromosomes by identifying their chromosomal position given in TIGR rice database. The genome organization and map location were investigated by the corresponding genome sequence with the map viewer at NCBI (<http://www.ncbi.nlm.nih.gov/mapview/>). The molecular weights and isoelectric points of OsSOTs were predicted by the Compute pI/Mw tool program (http://cn.expasy.org/tools/pi_tool.html). Genes separated by ≤ 5 genes were considered to be putative tandem duplicates. Such putative duplicates were analysed using online Vista Tools for Comparative Genomics (Frazer et al., 2004) for further analysis.

Phylogenetic analysis

The amino acid sequences of OsSOTs and AtSOTs were initially aligned using the program ClustalW (ver 1.83) and computing pairwise distance then discarding At3g51210 protein sequence. The unrooted phylogenetic tree was constructed by the Neighbor-Joining method and displayed using the same

software MEGA (version 5.02). Bootstrap analysis was performed using 1000 replicates, p-distance Method, pairwise deletion gaps data treatment.

Motif identification

We identified the protein motifs of two types of OsSOT genes using MEME (<http://meme.sdsc.edu/meme/meme.html>) with the motif length set at 3–300, motif sites 20, and the maximum number of motifs to find was defined as 20. The results were then manually adjusted. Furthermore, we characterized the two motifs using DNAMAN (version 4.0).

Promoter analysis

DNA sequences of rice SOT genes and 1500 bp ahead of the translation initiation codon (ATG) were collected from the GRAMENE database. The promoters of *OsSOTs* were analyzed by Plant-CARE database, and validated the results using the PLACE databases.

In silico expression analysis of OsSOT genes

cDNA signatures from Parallel Signature Sequencing (MPSS) and EST abundance data from NCBI were used to detect the expression patterns of rice PP2C genes in different organs. Expression evidence from Massively Parallel Signature Sequencing (MPSS) tags was determined from the Rice MPSS project (Nakano et al., 2006). MPSS data for 17-base signatures from 32 mRNA (except *OsSOT11*, *OsSOT29* and *OsSOT31*) libraries representing different plant growth and development, tissues/organs and stress conditions of rice were used for the analysis. Data for 17-base significant signatures, which uniquely identify an individual gene using the locus ID for sulphate transporters given in the TIGR database and show a perfect match (100% identity over 100% length of the tag), were selected and analyzed. The normalized abundance of these signatures for a given gene in these libraries represents a quantitative estimate of expression of that gene in terms of transcripts per million.

Microarray data analysis

Different groups had carried out microarray analyses on rice plants and CEL files for these were publicly available in the GEO database. We had used the GEO database for expression data for reproductive development (GSE6893 and GSE7951), expression data for stress treatment (GSE6901), expression data for auxin and cytokinin response (GSE5167), expression data for cytokinin response (GSE6719), our analysis on different reproductive developments for various stress treatments was used as well for expression analysis of the rice SOT gene family. The Affymetrix CEL files of each were imported and analyzed using Gene Chip Robust Multi-Array method. The IDs of probe sets present on the Affymetrix rice genome array representing the SOT genes were identified using the Rice Multi-platform Microarray Search tool (Saeed et al., 2003).

Data for all the probe sets for each SOT gene were used for expression analysis to study whether transcripts for some of the splice variants were differentially accumulated. After normalization of data for all the rice genes present on the chip, the log signal intensity values for rice probe IDs corresponding to SOT genes were extracted as a subset for subsequent analysis. Cluster analysis on rows was using log transformation performed, Average Linkage rule of Hierarchical clustering.

Results

Identification of genes for SOT proteins in rice

The key word of sulfotransferase search resulted in a total

of 32 sequences from TIGR pseudomolecule version 6.1 and 35 sequences were obtained from GRAMENE by the same method. Also the BLAST at NCBI resulted in 18 homologous genes. According to the SMART and PFAM Database analysis, all 35 proteins were found containing sulfotransferase domain. Taken together, a total of 35 non-redundant gene loci were predicted in rice. Almost all the OsSOT gene transcripts were supported by their full length cDNA (Table 1), the Table 1 showed an overview of the complete rice SOT family with additional information including gene identification, numbers of amino acids, ORF and genomic sequences, molecular weight and isoelectric point of proteins. Majority members of this family consist of an approximately number of amino acids of at least 330 that was similar to those SOTs in Arabidopsis.

Table 1. Features of the SOT family from rice and localization prediction for the SOT proteins.

Gene name	Gene ID (GRAMENE and TIGR)	Gene length (bp)	cDNA length (bp)	No. of introns	ORF length (bp)	Protein length (aa)	Mol. wt. (kDa)	pI	EST (NCBI)	Localization prediction
OsSOT17	LOC_Os08g40330	1082	1050	1	1050	350	39.18	8.8	no item	M (0.593)
OsSOT18	LOC_Os08g40380	1200	1200	0	999	333	37.53	6.88	0	M (0.748)
OsSOT19	LOC_Os08g40390	976	864	1	864	288	31.67	7.05	no item	M (0.591)
OsSOT20	LOC_Os09g08190	1160	1160	0	843	281	31.92	7.32	5	M (0.640)
OsSOT1	LOC_Os01g20950	1407	1407	0	1041	347	38.71	5.26	2	M (0.454); C (0.450)
OsSOT4	LOC_Os04g44460	1041	1041	0	1041	347	38.22	5.75	1	M (0.526); CS (0.482)
OsSOT6	LOC_Os06g42120	1274	1274	0	1116	372	41.31	6.74	1	CS (0.863); MMS (0.620)
OsSOT23	LOC_Os11g04530	3021	1311	3	1311	437	48.91	7.26	no item	M (0.748)
OsSOT24	LOC_Os11g04540	1025	951	1	951	317	35.45	7.19	no item	M (0.588); CS (0.578)
OsSOT25	LOC_Os11g04550	1083	1017	0	1017	339	38.87	7.21	no item	M (0.748)
OsSOT31	LOC_Os12g04300	806	573	1	573	191	22.04	5.61	no item	M (0.748); MMS (0.615)
OsSOT32	LOC_Os12g04320	1257	1257	0	1014	338	38.39	6.88	5	M (0.640)
OsSOT33	LOC_Os12g04340	1204	1204	0	867	288	33.09	7.14	7	O (0.370); M (0.269)
OsSOT34	LOC_Os12g04350	1275	873	1	873	291	32.89	6.57	no item	M (0.748)
OsSOT2	LOC_Os02g45890	1378	1378	0	1077	359	39.5	6.83	1	M (0.550)
OsSOT21	LOC_Os09g38239	1502	1502	0	1158	386	41.71	6.64	15	M (0.479); C (0.450)
OsSOT27	LOC_Os11g26390	1174	1174	0	1011	337	38.22	6.14	no item	C (0.450); M (0.430)
OsSOT28	LOC_Os11g30810	1847	1255	1	987	329	36.01	6.51	5	M (0.463); N (0.300)
OsSOT29	LOC_Os11g30830	543	543	0	543	180	20.07	6.24	no item	M (0.550)
OsSOT30	LOC_Os11g30910	1756	1756	0	1188	396	43.42	7.15	no item	M (0.673); N(0.600)
OsSOT7	LOC_Os07g05450	1383	579	2	579	193	21.48	6.5	no item	C (0.450); M (0.307)
OsSOT8	LOC_Os07g05460	1730	1469	1	1062	354	39.36	7.82	0	N (0.700)
OsSOT12	LOC_Os08g14180	4598	1269	2	1269	423	45.59	7.99	1	C (0.450); CS (0.400)
OsSOT13	LOC_Os08g14190	2968	1234	1	1047	349	38.23	7.83	2	C (0.450)
OsSOT14	LOC_Os08g17510	1354	570	1	570	190	21.16	4.78	no item	C (0.450); M (0.389)
OsSOT16	LOC_Os08g20130	1589	1315	1	1008	336	37.32	6.08	6	M (0.530)
OsSOT35	LOC_Os12g17160	2391	1948	1	1017	339	37.57	6.73	43	N (0.960); M (0.451)
OsSOT5	LOC_Os06g14220	1258	1258	0	615	205	22.22	8.52	2	C (0.650)
OsSOT9	LOC_Os07g41460	1469	1469	0	1032	344	37.78	6.61	18	M (0.657)
OsSOT11	LOC_Os07g41560	535	435	1	435	145	15.81	5.9	no item	C (0.450); M (0.300)
OsSOT15	LOC_Os08g17520	984	669	1	669	223	23.98	8.49	no item	N (0.300);CS (0.200)
OsSOT10	LOC_Os07g41540	1341	585	2	585	195	21.92	4.95	no item	M (0.640)
OsSOT22	LOC_Os10g11270	2403	1291	1	1104	368	41.03	6.12	2	M (0.563)
OsSOT26	LOC_Os11g15370	2619	882	1	882	294	32.19	10.89	no item	N (0.700); M (0.481)
OsSOT3	LOC_Os04g29000	2267	1461	0	1089	363	40.9	5.86	3	C (0.450);M (0.357)

The program PSORT was used for the localization prediction (<http://psort.ims.u-tokyo.ac.jp>). Abbreviations: bp, base pair; aa, amino acids; ID, identification; No., number.

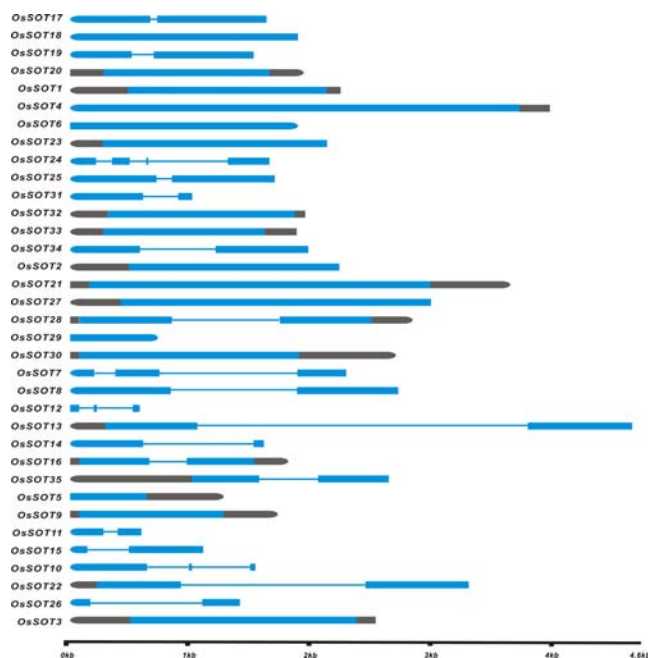


Figure 1. Exon-intron structure of alternatively spliced OsSOT genes. The gene names are mentioned on the left, all 35 genes are arranged order to which subfamily they belonging. The blue bars represent the exons, the lines represent the introns while the gray represent unencoded Non-coding region.

The presence of an existing cDNA with a known corresponding position on the genome was required for the determination of the number and length of introns. Based on these sequences, the map of introns was drawn as in Fig. 1: 16 of 35 genes did not contain any introns, 15 had one intron, 3 had two introns and only one *OsSOT23* had three introns.

Chromosomal localization and Tandem duplications

The 35 SOT genes were localized on the 10 chromosomes of rice by virtue of their presence on the TIGR pseudomolecules and GRAMENE (Fig. 2). It was predicted that the formation of the one gene pair, *OsSOT25* and *OsSOT32*, might be due to the segmental duplication of the rice genome. SOT genes were scattered all over and in clusters (Fig. 2), even at the telomeric and centromeric regions. A maximum number of 8 genes were present on chromosome 8 and 11 followed by 5 on chromosome 12. On the other hand, none gene was present on chromosome 3 and 5. And mostly SOT genes being transcribed in the reverse orientation, *OsSOT20* and *OsSOT21* were present on chromosome 9 and transcribed in the forward orientation. *OsSOT16* and *OsSOT20* were localized very near the centromere.

The localization of SOTs on the chromosomes depicted the clustering of sixteen genes on the four chromosomes. A total of six clusters of tandemly arranged SOT genes were observed on different chromosomes. Chromosome 7 harbored one cluster of SOT genes, first cluster of 2 members of D subfamily. The second cluster of 2 members of D subfamily and third cluster of 3 members of A subfamily were present on chromosome 3. The fourth cluster of 3 members of C subfamily and fifth cluster of 2 members of B subfamily were localized on chromosome 11. The last cluster of 4 members of B subfamily was on chromosome 12.

Phylogenetic analysis of SOT family in rice

Using the 35 SOT full length amino acid sequences from rice, a phylogenetic analysis was performed to construct a phylogenetic tree (Fig. 3). The results showed that all but 2 OsSOTs could be divided into 7 subfamilies. 7 subfamilies were identi-

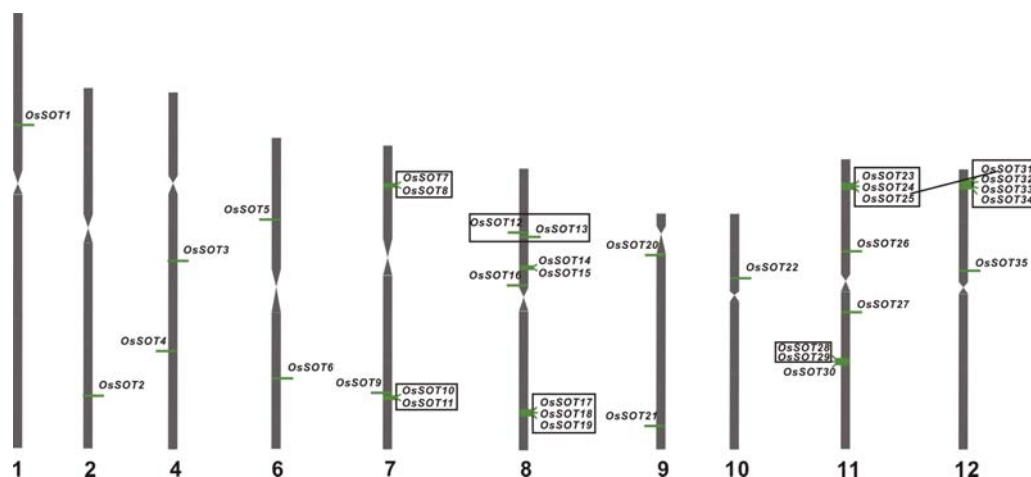


Figure 2. Chromosomal distribution of rice OsSOT genes. The chromosome number is indicated at the bottom of each chromosome. OsSOT genes localized on duplicated chromosomal segment were connected by line. The genes labeled on left side were oriented from bottom to top and those on right side were oriented from top to bottom on rice chromosomes. Six clusters of tandemly arranged OsSOT genes were indicated in boxes. Exact position of each OsSOT gene on rice chromosome pseudomolecules was given in Supplementary Table S1.

Table 2. Putative motifs predicted in rice SOT proteins.

Motif	e-value	sites	length(aa)	Sequences
1	2.0e-623	30	41	WxHYLGYWxxSxxxPxxVLFLxYEELLxDPxxxVRKLAEFx
2	5.7e-470	29	29	NxxFFRKGxxGDWxNHMTPEMAxRLDxIx
3	2.8e-472	28	29	PRxxDVLLATFPKSGTTWLKALxLATxxR
4	4.6e-488	30	40	GxPFTxEEEEAGVVxxIVELCSFExLxxLEVNxxG
5	2.6e-319	27	21	GCKVVYLCRDPKDxLVSLWHF
6	1.4e-257	29	21	DLDSLPSRLxTHMPxSLLP
7	1.4e-241	30	19	HPLRLNPHDCVPFLExxF
8	3.9e-76	9	21	PYSRHPDGWYFTxGVVSAMV
9	9.7e-82	20	21	RxYxGxWxxExxAxGxxAxRR
10	1.4e-59	5	41	MAPSFPLSSAPxxADETAHKEIYDQLRQxVETFTAPSxI
11	1.7e-15	4	29	GPVAFKDAVDVDAxPVRPxxTEHDAAxSAM

Significant motifs (e-value <e-100) of more than 10 amino acid length were predicted by MEME search. The consensus sequence, length (amino acids), number of OsSOT proteins containing the motif and e-value of each predicted motif is given.

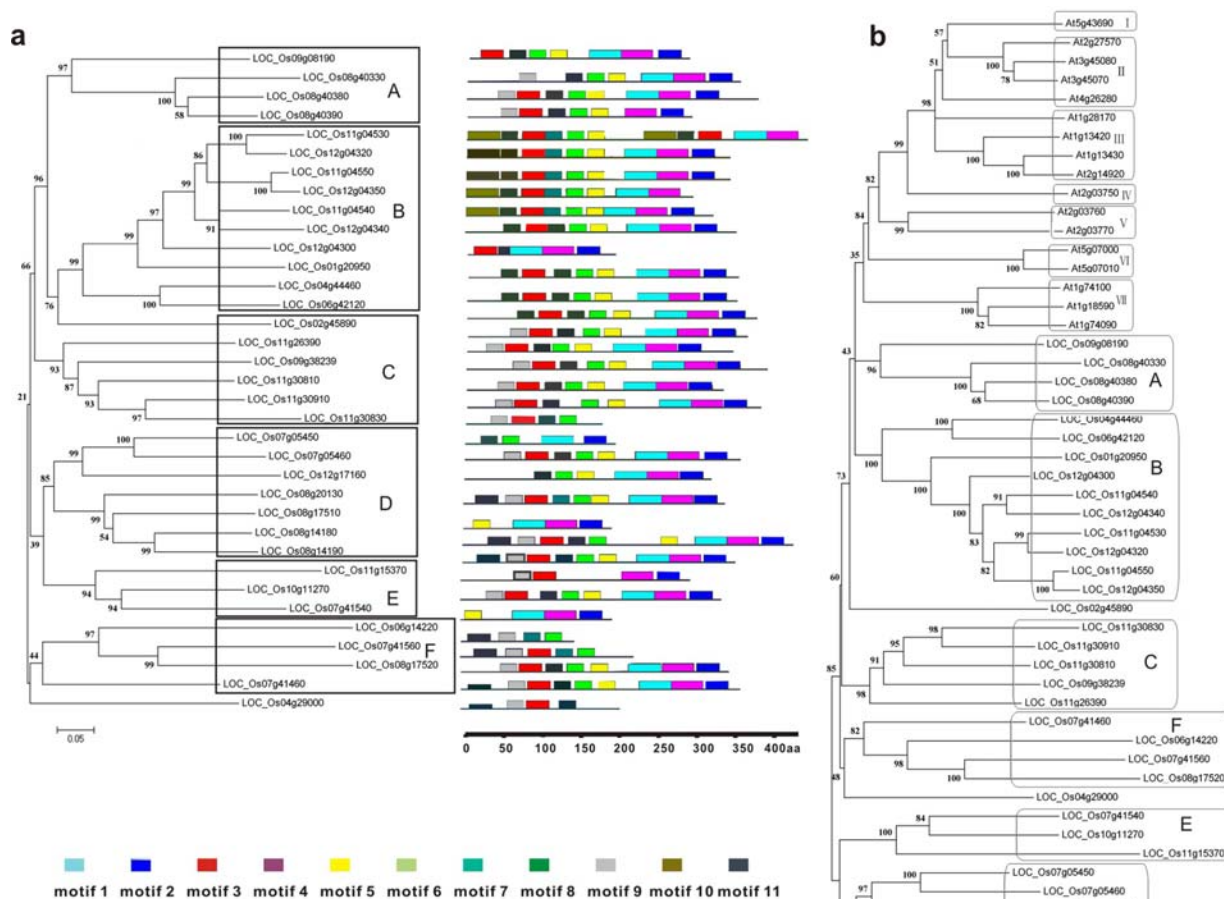


Figure 3. An analytical view on the SOT gene family in rice and the phylogenetic relationships of Arabidopsis and rice. A. Left: The unrooted tree was constructed using aligned full-length amino acid sequences and summarizes the phylogenetic relationship among the 35 members of SOT family in rice. The tree shows the 7 major phylogenetic subfamilies marked with different alternating tones of a square. Right: Each colored box with a number represents a specific motif in the protein identified using the MEME motif search tool. The order of the motifs corresponds to their position within individual protein sequences. B. The unrooted distance tree was inferred using the neighbor-joining method and is based on an alignment of the full-length amino acid sequences of the 17 Arabidopsis and 35 rice genes.



opmental stages available at the GEO database under accession numbers GSE6893 and GSE7951, respectively. We studied the expression of different rice sulfotransferases using microarray data available online to analysis the gene expression a total of 32 *OsSOT* genes were represented on the Affymetrix arrays (Fig. 6). There was no probe set on the arrays corresponding to the remaining *OsSOT* genes.

Though the overall mostly *OsSOT* genes were expressed exclusively in stigma and ovary. 30 genes (without *OsSOT24* and *OsSOT34*) passed data filter. 17 *SOT* genes were expressed high in root. All the *OsSOT* genes were expressed lowly in shoot apical meristem (SAM) and young leaf (YL). Also mostly *OsSOT* genes were expressed lowly in mature leaf (ML), except *OsSOT9* was expressed high. 8 *OsSOT* genes were preferentially expressed at P5 stage of panicle development and the expression of 3 *OsSOT* genes were high

during P6 stage. The expression of *OsSOT12* was restricted to late stages of panicle development (P5–P6) and seed development stages (S1–S4). These genes may perform specific roles in these tissues/organs or developmental stage.

Among the three *OsSOT* genes of the fourth cluster localized on duplicated chromosomal segments, one pair *OsSOT23/25* exhibit similar expression patterns. All the other five clusters of tandemly duplicated *OsSOT* genes exhibited similar to highly diverged expression patterns. It was interesting that two pairs *OsSOT7/8* and *OsSOT12/13*, among them *OsSOT8* and *OsSOT12* were expressed lowly in stigma and ovary, but *OsSOT7* and *OsSOT13* were expressed high (Fig. 6).

Differential expression of *SOT* genes during hormone treatment

First dataset included microarray analysis of root and leaf

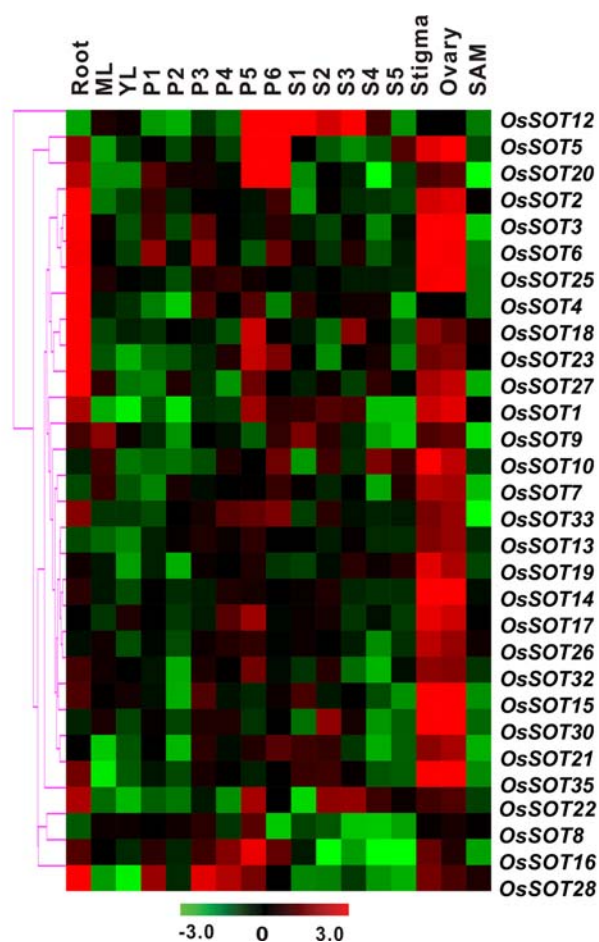


Figure 6. Expression patterns of rice *SOT* in various tissues and developmental stages. Hierarchical clustering analysis of 32 *SOT* genes represented on Affymetrix Rice Genome Array is shown. For clustering we used average $\log_2$ signal values for three biological replicates of each sample after normalization of the raw data. The color scale for log signal values is shown at the bottom. ML, mature leaf; YL, young leaf; SAM, shoot apical meristem; P1–P6, stages of panicle development; S1–S5, stages of seed development.

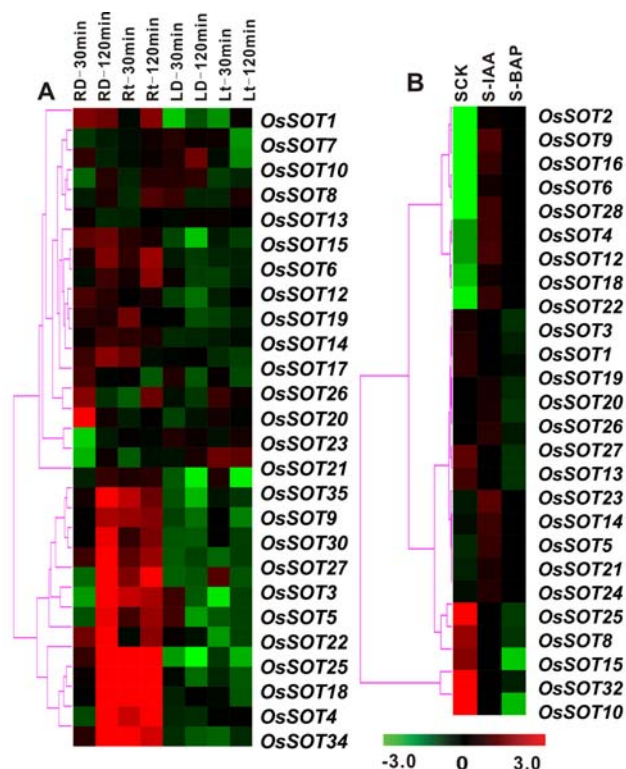


Figure 7. Differential *OsSOT* expressions at hormone treatment. A. Two-week-old seedlings treated tZ and DMSO for 30 min and 120 min; B. 7-day-old rice seedlings treated with IAA and BAP. For clustering we used average $\log_2$ signal values for three biological replicates of each sample after normalization of the raw data. The color scale for log signal values is shown at the bottom. RD, root tissues of two-week-old seedlings treated with dimethyl sulfoxide; Rt, root tissues of two-week-old seedlings treated with trans-zeatin; LD, leaf tissues of two-week-old seedlings treated with dimethyl sulfoxide; Lt, leaf tissues of two-week-old seedlings treated with trans-zeatin; SCK, 7-day-old rice seedlings; S-IAA, 7-day-old rice seedlings treated with indole-3-acetic acid; S-BAP, 7-day-old rice seedlings treated with benzyl aminopurine.

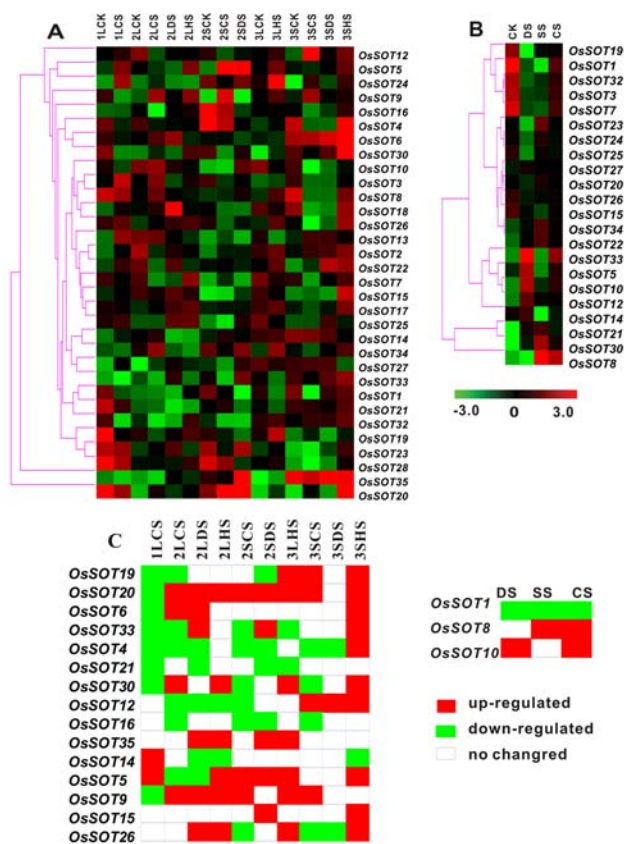


Figure 8. Differential rice *SOT* expressions in response to various abiotic conditions. Hierarchical clustering of *SOT* genes showing differential expression under abiotic stress condition is shown. A. Differential developmental stages under various abiotic conditions; B. 7-days old young seedling treated with abiotic stress condition; C. The genes up-regulated or down-regulated by more than 2-fold in any of the three abiotic stresses given to as compared with CK have been shown. 1, seedling stage; 2, booting stage; 3, heading and flowering stage; L, leaf; S, spike; CS, cold stress; DS, dehydration stress; HS, high temperature stress; SS, salt stress; CK, as compared with an untreated seedling. The color bar representing $\log_2$ signal values is shown at the bottom.

tissues of two-week-old seedlings treated with trans-zeatin (tZ) and dimethyl sulfoxide (DMSO) for 30 min and 120 min (Fig. 7A). Second dataset included microarray analysis of 7-day-old rice seedlings treated with indole-3-acetic acid (IAA) and benzyl aminopurine (BAP), just 26 genes passed data filter (Fig. 7B). The data analysis showed that a total of 26 *OsSOT* genes exhibited significant differential expression under at least one of the two conditions analyzed. Four genes (*OsSOT8/10/15/32*) which showed low expression and nine genes (*OsSOT2/4/6/9/16/12/18/22/28*) showed significant expression in the presence of BAP and IAA. Mostly genes were expressed high in root and lowly in leaf after tZ and DMSO treatment for 120 min compared to for 30 min. After BAP and IAA treatment, of the pair *OsSOT7/8*, *OsSOT8* gene showed large down-regulated expression (13.7

and 24.26 fold) but *OsSOT7* did not. Three genes, *OsSOT16* (76.89 and 46.32 fold) and *OsSOT28* (35.08 and 19.47 fold) exhibit significant up-regulated expression in the presence of IAA and BAP, and *OsSOT22* was up-regulated (16.58 and 10.02 fold).

Expression of sulfotransferase gene family members during abiotic stresses

The microarray dataset available for dehydration, salt, high temperature and cold stresses also showed differential expression pattern for the *OsSOT* gene family in differential developmental stages (Fig. 8A), all the 32 genes passed data filter. Differential expression analysis for rice 7-days old seedlings treated with different abiotic stresses (dehydration, salt and cold) as compared to mock-treated control seedlings was performed (Fig. 8B), 22 genes passed data filter. Microarray analysis performed on seedling, booting, heading and flowering stages subjected to cold, dehydration and high temperature stress, revealed fifteen genes 2-fold down- or up-regulated genes in at least one of the stress conditions. The expression of members of sub-family A, *OsSOT20* was up-regulated in leaf and spike at booting, heading and flowering stages in response to three stresses; but *OsSOT19* was down-regulated in leaf and spike at seedling and booting stages in response to cold and drought stresses. *OsSOT20* was up-regulated in spike at heading and flowering stage in response to three stresses, and the level of expression up to 182.12 fold. *OsSOT14* was just down-regulated in leaf and spike at booting stage in response to drought stress, *OsSOT35* up-regulated in leaf at heading and flowering stage to high temperature stress. The members of sub-family F, *OsSOT5/9/15* were up-regulated in 2S-CS, 2S-DS or 3S-HS (Fig. 8C). However, microarray analysis performed on 7-days old seedling just different three genes (*OsSOT1*, *OsSOT8* and *OsSOT10*) down-regulated or up-regulated subjected to any stress. *OsSOT1* was down-regulated to all the three stresses (Fig. 8C).

The promoter of abiotic stress responding genes analyzed

To investigate the probable explanation for abiotic stress responding genes, we analyzed their promoter sequences about 1.5 kb upstream of translational start site. The results of Plant-CARE database and the PLACE databases shown that a total of 21 types of *cis*-regulatory elements (CREs) were identified responding to abiotic stress, which did not contain the common eukaryotic regulatory elements, such as TATA and CAAT boxes. The names and functions of these 21 motifs were shown in Table 3. Their amounts obtained by the Plant-CARE database were displayed in Supplementary Fig. S1. The motifs, ARE, ABRE and TGACG-motif were found in 7 out 18 genes. In reverse, the motifs MBS I, MRE, TATC-box, P-box, C-repeat/DRE, GC-motif, AuxRR-core, ATGCAAAT motif, CCAAT-box were just found in one gene respectively. Other motifs distributed in 4 or 5 genes, apart

Table 3. Putative regulatory *cis*-element sequences in 5'-upstream regions of 11 *SOTs* in rice.

<i>cis</i> -regulatory elements	Function
5UTR Py-rich stretch	<i>cis</i> -acting element conferring high transcription levels
ARE	<i>cis</i> -acting regulatory element essential for the anaerobic induction
CGTCA	<i>cis</i> -acting regulatory element involved in the MeJA-responsiveness
MBS	MYB binding site involved in drought-inducibility
O2-site	<i>cis</i> -acting regulatory element involved in zein metabolism regulation
ABRE	<i>cis</i> -acting regulatory element involved in the ABA-responsiveness
GARE-motif	gibberellin-responsive element
TGACG-motif	<i>cis</i> -acting regulatory element involved in the MeJA-responsiveness
LTR	<i>cis</i> -acting element involved in low-temperature responsiveness
MBS I	MYB binding site involved in flavonoid biosynthetic genes regulation
MRE	MYB binding site involved in light responsiveness
TATC-box	<i>cis</i> -acting element involved in gibberellin-responsiveness
TC-rich repeats	<i>cis</i> -acting element involved in defense and stress responsiveness
TCA-element	<i>cis</i> -acting element involved in salicylic acid responsiveness
P-box	gibberellin-responsive element
C-repeat/DRE	regulatory element involved in cold- and dehydration-responsiveness
GC-motif	enhancer-like element involved in anoxic specific inducibility
HSE	<i>cis</i> -acting element involved in heat stress responsiveness
AuxRR-core	<i>cis</i> -acting regulatory element involved in auxin responsiveness
ATGCAAAT motif	<i>cis</i> -acting regulatory element associated to the TGAGTCA motif
CCAAT-box	MYBHv1 binding site

from 5UTR Py-rich stretch in *OsSOT9* and *OsSOT16*. *OsSOT4* contained 8 CREs, *OsSOT5* and *OsSOT16* had 7, other genes contained about 4-6 motifs.

Discussion

Thirty-five SOT in rice had been identified, however the understanding of genetic and molecular was not clear (Cinta and Luc, 2008). In this study we analyzed the 35 OsSOT genes and found that majority members of this family consist of an approximately number of amino acids of at least 330 that was similar to SOT in Arabidopsis, most of the rice SOT genes contain no introns or one in contrast to Arabidopsis SOT genes nearly all contain no introns. Genes for human SOTs, as well as for other mammalian SOTs, show a high degree of structural homology with conservation of the locations of most intron/exon splice junctions (Klein and Papenbrock, 2004).

The 35 SOT genes could be localized on the 10 chromosomes of rice, and six clusters of tandemly arranged SOT genes, just one gene pair segmental duplication (Fig. 2). This data suggested the major contribution of tandem duplications to the SOT family expansion. On the other hand, we found just two members of SOT family located on the segmental duplicated region. Since the number of SOT genes located on segmental duplicated regions was much smaller than those present in tandem, the localized gene duplications appear to have contributed a major role in expansion of this gene family.

In order to evaluate the evolutionary relationship among the

35 rice SOT proteins, we also performed a phylogenetic analysis based on their full length amino acid sequences (Fig. 3A). We further investigated the relationship between the Arabidopsis and rice SOT protein families by generating an alignment of the full-length protein sequences of 35 OsSOTs and 17 AtSOTs followed by the construction of a phylogenetic tree (Fig. 3B). The rice and Arabidopsis genes in this family were very not closely related to each other, within each subfamily no clear orthology relationships emerge between them, indicating independent evolution by duplication (or gene loss) in the two species.

11 highly significant putative conserved motifs were identified. The different motif distribution in the respective protein sequences may provide an anchor to study the divergence of gene function in different subfamilies. Furthermore, the fact that proteins in the same subfamily exhibit highly similar motif distribution supports their close evolutionary relationships. All OsSOTs had conserved amino acid motifs which were involved in PAPS binding regions I and IV, however not all the OsSOTs had the two domains, besides 10 had one domain, other 25 had both the two regions. Compared to 14 of 18 AtSOTs contained two conserved amino acid motifs (Klein and Papenbrock, 2004). This suggested OsSOT protein is more discrepant during evolution progress.

In mammalian species one group of SOT proteins was associated with membranes, and a second group of SOT proteins was soluble (Sugahara et al., 2003). In plant, the size of this membrane SOT protein was 42 kDa, whereas most soluble plant SOT proteins have only a molecular mass of about 35

kDa. The difference might reflect the addition of a transmembrane domain (Varin et al., 1997). OsSOT proteins were mostly about 35 kDa and several 42 kDa (Table 1). OsSOT25 was suggested as peroxisomal by the PTS1 predictor, though most SOT proteins transport into peroxisome with probabilities higher. But the computer predictions for the intracellular localization were not revealed clear and reliable results. Therefore, the intracellular localization has to be investigated experimentally (Bauer et al., 2004; Nowak et al., 2004).

The mRNA levels for most of the *Arabidopsis* SOTs characterized to date were low and reflected by the relatively low number or even no hits of EST clones, with the exception of *AtSOT15* and *AtSOT16*. Expression signatures for 32 OsSOT genes were detected in the MPSS database, and 17 members had matching ESTs. No expression evidence was found in any database for six OsSOT genes, suggesting that they might be pseudogenes.

Gene expression patterns can provide important clues for gene function. To date, a comparative analysis of the expression patterns of the whole SOT gene family in rice has not been performed. The microarray data indicated that all the OsSOT genes were expressed at not high level and exhibit various expression patterns. All the OsSOT genes were expressed lowly in shoot apical meristem (SAM) and young leaf (YL) and mostly OsSOT genes were expressed lowly in mature leaf (ML), but very high in stigma and ovary, showed that OsSOT genes with respect to age of tissue-type. Consistent with *AtST1* /*At2g03770* under normal physiological conditions, expression at the steady-state mRNA level was highest in the aerial part of seedlings (Lacomme and Roby, 1996). *AtST4* subfamily members were found to be largely root specific (Walia et al., 2005), for OsSOT genes, 17 expressed at a high level in root. Because the expression pattern of a gene could be indicative of its functional relevance, the wide expression patterns of some OsSOT family members might play regulatory roles at multiple development stages while the unique expression patterns of other members might indicate these genes participate in specific biological processes. The divergent expression patterns of 3 cluster gene pairs suggesting that splicing of these SOTs were tissue specific (Fig. 6).

There were reports that over-expression of *AtSOT16* may lead to increased production of IAA and elemental sulfur in roots (Klein and Papenbrock, 2008), the observed pattern of *BNST3* promoter activity was first characterized as a gene up-regulated in response to pathogens and the pathogen-related signals, methyl jasmonate, and salicylic acid (Marsolais et al., 2004). *BNST* genes were similarly inducible by salicylic acid (Lacomme and Roby, 1996), were also induced in response to various forms of chemical stress, including ethanol and other xenobiotics, and the herbicide safer naphthalic anhydride (Marsolais et al., 2004). To study modulation in the expression of members of the rice SOT gene fam-

ily by plant hormones, we used two microarray datasets to assess the effect of IAA and BAP on the expression profiles. The data analysis showed that different members of this gene family were differentially regulated by these plant hormones and with respect to age of seedlings, tissue-type and/or cytokinin type (Fig. 7).

Several *AtSOT* had been study, such as *AtSOT12*/ *At2g03760*, gene expression was strongly induced by salt, osmotic stress and hormone treatments and over-expressing plants were more resistant to pathogen infection. *AtSOT12* mRNA was induced also by salicylic; *AtSOT17* /*18* were involved in plant defense and response to abiotic stresses (Baek et al., 2010; Piotrowski et al., 2004). The results of expression of sulfotransferase gene family members during abiotic stresses indicated that 11 OsSOT genes were response to one abiotic stress at least (Fig. 8C). Meantime the members of gene cluster had not same expression pattern. For genes form the same subfamily, for example, *OsSOT19* was down-regulated while *OsSOT20* was up-regulated in response to cold and drought stresses, also were different. But *OsSOT5/9/15* of sub-family F, were up-regulated in 2S-CS, 2S-DS or 3S-HS. This might because the members of same subfamily had different or no overlap function, similar to *AtSOTs*, such as *AtSOT12* gene was clustered with and encoded a protein with high sequence similarity to *AtSOT11*/*At2g03750* and *AtSOT13*/ *At2g03770*, however, mutation in *AtSOT11* and *AtSOT13* did not cause a more sensitive phenotype of the mutants to SA (Baek et al., 2010). The results of promoter sequences of abiotic stress responding genes showed that these genes contained *cis*-regulatory elements with responding to abiotic stresses. *OsSOT20* was up-regulated induced by dehydration, high temperature and cold stresses, and its promoter contained 5UTR Py-rich stretch, ARE, CGTCA and MBS motifs that indicated the gene maybe induced by JA and abiotic stress. The considerable difference in the regulatory elements of same subfamily genes might explain their divergent expression patterns. However, experimental validation is required to reach this conclusion. Further, the existence of some other regulatory mechanism, which is responsible for divergent expression patterns and/or non-functionalization of the genes, could not be ruled out.

Acknowledgements This research was supported by Fund 11ZA090 (A Project Supported by Scientific Research Fund of Sichuan Provincial Education Department) to RJ Chen. We thank Dr Xie Tan and Pedro for critical reading of this manuscript.

References

- Ananvoranich S, Varin L, Gulick P and Ibrahim R (1994) Cloning and regulation of flavonol 3-sulfotransferase in cell-suspension cultures of *Flaveria bidentis*. *Plant Physiol.* 106: 485–491.
- Baek D, Pathange P, Chung JS, Jiang J, Gao L, Oikawa A, Hirai

- MY, Saito K, Pare PW and Shi H (2010) A stress-inducible sulphotransferase sulphonates salicylic acid and confers pathogen resistance in *Arabidopsis*. *Plant, Cell Environ.* 33: 1383-1392.
- Bauer M, Dietrich C, Nowak K, Sierralta WD and Papenbrock J (2004) Intracellular localization of *Arabidopsis* sulfurtransferases. *Plant Physiol.* 135: 916-926.
- Cinta HS (2008) Sulfotransferases from Plants, Algae and Phototrophic Bacteria. *Sulfur Metabolism in Phototrophic Organisms*, Cinta HS and Luc V, eds., Academic Press, Montreal, CA, pp. 111-130.
- Faulkner I and Rubery P (1992) Flavonoids and flavonoid sulphates as probes of auxin-transport regulation in *Cucurbita pepo* hypocotyl segments and vesicles. *Planta* 186: 618-625.
- Frazer KA, Pachter L, Poliakov A, Rubin EM and Dubchak I (2004) VISTA: computational tools for comparative genomics. *Nucleic Acids Res.* 32: W273-W279.
- Gidda S and Varin L (2006) Biochemical and molecular characterization of a flavonoid 7-sulfotransferase from *Arabidopsis thaliana*. *Plant Physiol. Bioch.* 44: 628-636.
- Gidda SK, Miersch O, Levitin A, Schmidt J, Wasternack C and Varin L (2003) Biochemical and molecular characterization of a hydroxyjasmonate sulfotransferase from *Arabidopsis thaliana*. *J. Biol. Chem.* 278: 17895-17900.
- Hempel F, Bozarth A, Sommer MS, Zauner S, Przyborski JM and Maier UG (2007) Transport of nuclear-encoded proteins into secondarily evolved plastids. *Biol. Chem.* 388: 899-906.
- Jacobs M and Rubery PH (1988) Naturally occurring auxin transport regulators. *Science* 241: 346-349.
- Klein M and Papenbrock J (2004) The multi-protein family of *Arabidopsis* sulphotransferases and their relatives in other plant species. *J. Exp. Bot.* 55: 1809-1821.
- Klein M (2008) Sulfotransferases and Their Role in Glucosinolate Biosynthesis. *Sulfur Assimilation and Abiotic Stress in Plants*, Klein M and Papenbrock J, eds., Academic Press, Hannover, CA, pp. 149-166.
- Klein M, Reichelt M, Gershenzon J and Papenbrock J (2006) The three desulfoglucosinolate sulfotransferase proteins in *Arabidopsis* have different substrate specificities and are differentially expressed. *FEBS J.* 273: 122-136.
- Lacomme C and Roby D (1996) Molecular cloning of a sulfotransferase in *Arabidopsis thaliana* and regulation during development and in response to infection with pathogenic bacteria. *Plant Mol. Biol.* 30: 995-1008.
- Marsolais F, Boyd J, Paredes Y, Schinas AM, Garcia M, Elzein S and Varin L (2007) Molecular and biochemical characterization of two brassinosteroid sulfotransferases from *Arabidopsis*, *AtST4a* (At2g14920) and *AtST1* (At2g03760). *Planta* 225: 1233-1244.
- Marsolais F (2000) Chapter Fourteen Plant soluble sulfotransferases: Structural and functional similarity with mammalian enzymes. *Recent Adv Phytochem*, Marsolais F, Gidda SK, Boyd J and Varin L, eds., Academic Press, Quebec, CA, pp. 433-456.
- Marsolais F, Sebastia CH, Rousseau A and Varin L (2004) Molecular and biochemical characterization of *BNST4*, an ethanol-inducible steroid sulfotransferase from *Brassica napus*, and regulation of BNST genes by chemical stress and during development. *Plant Sci.* 166: 1359-1370.
- Nakamura Y, Mith fer A, Kombrink E, Boland W, Hamamoto S, Uozumi N, Tohma K and Ueda M (2011) 12-Hydroxyjasmonic Acid Glucoside Is a COI1-JAZ-Independent Activator of Leaf-Closing Movement in *Samanea saman*. *Plant Physiol.* 155: 1226-1236.
- Nakano M, Nobuta K, Vemaraju K, Tej SS, Skogen JW and Meyers BC (2006) Plant MPSS databases: signature-based transcriptional resources for analyses of mRNA and small RNA. *Nucleic Acids Res.* 34: D731-735.
- Nowak K, Luniak N, Meyer S, Schulze J, Mendel R and Hnsch R (2004) Fluorescent proteins in poplar: a useful tool to study promoter function and protein localization. *Plant Biol.* 6: 65-73.
- Nowell S and Falany C (2006) Pharmacogenetics of human cytosolic sulfotransferases. *Oncogene* 25: 1673-1678.
- Piotrowski M, Schemenewitz A, Lopukhina A, Müller A, Janowitz T, Weiler EW and Oecking C (2004) Desulfoglucosinolate sulfotransferases from *Arabidopsis thaliana* catalyze the final step in the biosynthesis of the glucosinolate core structure. *J. Biol. Chem.* 279: 50717-50725.
- Rouleau M, Marsolais F, Richard M, Nicolle L, Voigt B, Adam G and Varin L (1999) Inactivation of brassinosteroid biological activity by a salicylate-inducible steroid sulfotransferase from *Brassica napus*. *J. Biol. Chem.* 274: 20925-20930.
- Saeed A, Sharov V, White J, Li J, Liang W, Bhagabati N, Braisted J, Klapa M, Currier T and Thiagarajan M (2003) TM4: a free, open-source system for microarray data management and analysis. *BioTechniques* 34: 374-378.
- Sugahara K, Mikami T, Uyama T, Mizuguchi S, Nomura K and Kitagawa H (2003) Recent advances in the structural biology of chondroitin sulfate and dermatan sulfate. *Curr. Opin. Struct. Biol.* 13: 612-620.
- Varin L, Chamberland H, Lafontaine JG and Richard M (1997) The enzyme involved in sulfation of the turgorin, gallic acid acid 4-O-(β -D-glucopyranosyl-6'-sulfate) is pulvinilocalized in *Mimosa pudica*. *Plant J.* 12: 831-837.
- Walia H, Wilson C, Condamine P, Liu X, Ismail AM, Zeng L, Wanamaker SI, Mandal J, Xu J and Cui X (2005) Comparative transcriptional profiling of two contrasting rice genotypes under salinity stress during the vegetative growth stage. *Plant Physiol.* 139: 822-835.
- Yang H, Matsubayashi Y, Nakamura K and Sakagami Y (1999) *Oryza sativa* PSK gene encodes a precursor of phytosulfokine- α , a sulfated peptide growth factor found in plants. *Proc. Natl. Acad. Sci. USA* 96: 13560.